

D180

Educational Research

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher writing a literature review on vocabulary instruction finds four sources. A popular teaching blog with 200,000 followers is the easiest to read. Which is the strongest source for the review?

- A.** The popular teaching blog.
- B.** A trade magazine article, since magazines always check facts.
- C.** A peer-reviewed journal article reporting a study's methods and results.
- D.** A vendor white paper about its vocabulary program.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A graduate student reads a well-designed textbook chapter, with colorful charts, that describes three studies on peer tutoring. Her instructor has asked for primary sources. What should she do?

- A.** Cite the chapter as a primary source.
- B.** Cite the charts, since they show the data.
- C.** Locate and cite the three original studies.
- D.** Cite a news story about the chapter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

District data from 40 schools show that schools with more library books have higher reading scores. The district just finished a new website. A board member concludes that buying more books will raise scores. What is the problem with this conclusion?

- A.** A link in existing data does not show cause; another factor such as school funding could explain both.
- B.** The sample of 40 schools is too small for any conclusion.
- C.** Higher scores cause schools to buy more books, so the reverse is proven.
- D.** Nothing; a strong relationship proves the books help.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A researcher wants to test whether a new homework routine causes higher math scores. Which study would support that causal claim?

- A.** Survey classes and correlate homework time with scores.
- B.** Randomly assign classes to use the routine or not, then compare scores.
- C.** Test one class before and after the routine.
- D.** Interview teachers who like the routine.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A teacher with a bright, well-decorated classroom uses a new spelling game from September to May. Her 24 students' scores rise 15 points. She concludes the game caused the gain. What weakens her claim most?

- A.** Nothing; any gain after a new method proves it worked.
- B.** There is no comparison group, so normal growth over the year could explain the gain.
- C.** The classroom decorations may have caused the gain.
- D.** 15 points is too small to matter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which of these designs gives the weakest basis for saying an intervention caused an improvement?

- A.** One group measured only after the intervention.
- B.** One group with a pre-test and a post-test.
- C.** Two similar groups, one receiving the intervention, with pre- and post-tests.
- D.** Random assignment to treatment and control groups.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A candidate's draft literature review, written over a busy spring break, has 12 paragraphs, each describing one article in order of publication date. What would most improve it?

- A.** Reorganize by themes, comparing where studies agree, conflict, and leave gaps.
- B.** Add more articles in the same format.
- C.** Cut to 12 sentences, one per article.
- D.** Reorder the paragraphs alphabetically by author.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A paragraph reads: “Three studies found feedback timing mattered for younger students, but two found no effect for older students, and none studied online classes.” What is this paragraph doing?

- A.** Proving feedback timing works for all students.
- B.** Summarizing a single study.
- C.** Reporting the author's own data.
- D.** Synthesizing: comparing findings across studies and naming a gap.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A fourth-grade teacher who also sits on the district wellness committee tries a new discussion routine in her own class, collects data, reflects, and adjusts for a second cycle. What is the main purpose of this work?

- A.** To advise the wellness committee.
- B.** To improve practice in her own classroom.
- C.** To prove the routine works in every school.
- D.** To test a theory with a large random sample.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A state agency wants findings about a reading program that apply to all of its districts. Why is classroom action research a poor fit for this goal?

- A.** Action research takes too many cycles to finish.
- B.** Action research can only be done by university faculty.
- C.** Action research never uses data.
- D.** Action research studies one local setting and is not designed to generalize.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A middle school teacher with a new interactive whiteboard drafts research questions for an action research study. Which is most researchable?

- A.** "Why does the state set the testing calendar in April?"
- B.** "How can we close all achievement gaps in the country?"
- C.** "How does adding a 5-minute retrieval quiz at the start of class affect my students' unit test scores?"
- D.** "Is the whiteboard good?"

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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12.

A teacher plans to use her students' journal entries as data in a study. Parents signed permission forms, which were printed on school letterhead. One 10-year-old says she does not want her journal included. What should the teacher do?

- A.** Include it, since the parent already signed.
- B.** Include it without her name, which removes the need for assent.
- C.** Stop the whole study for all students.
- D.** Leave that student's journal out of the study data.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A teacher writing a literature review on vocabulary instruction finds four sources. A popular teaching blog ~~with 200,000 followers is the easiest to read~~. Which is the strongest source for the review?

Target word: strongest source type **Decides:** Peer review and reported methods.

A	Camouflage	Reach and readability do not make it scholarly.
B	Absolute language	Editing is not peer review of methods.
C	Correct	Expert-checked and transparent about methods.
D	Adjacent category	Not peer reviewed and the author has a stake.

2. Answer C Concept 2

A graduate student reads a well-designed textbook chapter, ~~with colorful charts~~, that describes three studies on peer tutoring. Her instructor has asked for primary sources. What should she do?

Target word: finding the primary source **Decides:** The chapter reports others' studies.

A	Confusion set	A chapter summarizing studies is secondary.
B	Camouflage	Charts in a summary are still secondary.
C	Correct	The original studies are the primary sources.
D	Inverted operation	Moves further from the original.

3. Answer A Concept 3

District data from 40 schools show that schools with more library books have higher reading scores. ~~The district just finished a new website~~. A board member concludes that buying more books will raise scores. What is the problem with this conclusion?

Target word: flaw in the causal claim **Decides:** The data show a correlation only.

A	Correct	Correlation cannot rule out third variables.
B	Wrong stem number	Sample size is not the main flaw; the design is.
C	Inverted operation	Reverse direction is possible but also not proven.
D	Absolute language	Strength of correlation does not show cause.

4. Answer B Concept 3 • transfer

A researcher wants to test whether a new homework routine causes higher math scores. Which study would support that causal claim?

Target word: design that can show cause **Decides:** Random assignment with a comparison group.

A	Confusion set	Correlation cannot show cause.
B	Correct	Random assignment balances other factors.
C	Dropped element	No comparison group.
D	Adjacent category	Opinions do not test effect.

5. Answer B Concept 4

A teacher ~~with a bright, well-decorated classroom~~ uses a new spelling game from September to May. Her 24 students' scores rise 15 points. She concludes the game caused the gain. What weakens her claim most?

Target word: weakness of a one-group design **Decides:** No comparison group over a school year.

A	Absolute language	Before-after gains alone do not show cause.
B	Correct	Without a control, maturation and teaching in general are not ruled out.
C	Camouflage	Decorations stayed the same; the flaw is the missing comparison.
D	Wrong stem number	The size of the gain is not the design flaw.

6. Answer A Concept 4 • transfer

Which of these designs gives the weakest basis for saying an intervention caused an improvement?

Target word: rank designs — reversed from strongest to weakest **Decides:** One group, measured only after.

A	Correct	No baseline and no comparison.
B	Adjacent category	Weak, but it at least has a baseline.
C	Confusion set	Has a comparison and baseline.
D	Inverted operation	That is the strongest design.

7. Answer A Concept 5

A candidate's draft literature review, ~~written over a busy spring break~~, has 12 paragraphs, each describing one article in order of publication date. What would most improve it?

Target word: structure of a review **Decides:** Article-by-article is summary, not synthesis.

A	Correct	Synthesis is organized by ideas across sources.
B	Dropped element	More summaries do not create synthesis.
C	Wrong stem number	Shorter summaries are still summaries.
D	Adjacent category	Still one article at a time.

8. Answer D Concept 5 • transfer

A paragraph reads: "Three studies found feedback timing mattered for younger students, but two found no effect for older students, and none studied online classes." What is this paragraph doing?

Target word: identify synthesis **Decides:** Compares studies and names a gap.

A	Absolute language	It notes mixed results.
B	Confusion set	It draws on five studies.
C	Adjacent category	It reviews others' findings.
D	Correct	Agreement, difference, and gap across sources.

9. Answer B Concept 6

A fourth-grade teacher ~~who also sits on the district wellness committee~~ tries a new discussion routine in her own class, collects data, reflects, and adjusts for a second cycle. What is the main purpose of this work?

Target word: purpose of action research **Decides:** Practitioner, own setting, cycles.

A	Camouflage	Her committee role is unrelated.
B	Correct	Action research aims at local improvement.
C	Absolute language	Action research does not aim to generalize.
D	Adjacent category	That describes traditional experimental research.

10. Answer D Concept 6 • transfer

A state agency wants findings about a reading program that apply to all of its districts. Why is classroom action research a poor fit for this goal?

Target word: limit of action research — reversed from its purpose **Decides:** Action research aims at local improvement, not generalization.

A	Adjacent category	Length is not the reason it cannot generalize.
B	Inverted operation	It is done by practitioners.
C	Absolute language	It relies on data; its scope is local.
D	Correct	Broad claims need representative samples and designs built to generalize.

11. Answer C Concept 7

A middle school teacher ~~with a new interactive whiteboard~~ drafts research questions for an action research study. Which is most researchable?

Target word: best research question **Decides:** Specific, measurable, within her control.

A	Adjacent category	Outside the teacher's control to study or change.
B	Absolute language	Too broad to study.
C	Correct	Names a specific change and a measure she controls.
D	Camouflage	Vague and centered on the tool detail.

12. Answer D Concept 8

A teacher plans to use her students' journal entries as data in a study. Parents signed permission forms, ~~which were printed on school letterhead~~. One 10-year-old says she does not want her journal included. What should the teacher do?

Target word: respecting assent **Decides:** The child declined assent.

A	Dropped element	Ignores the child's assent.
B	Adjacent category	Confidentiality does not replace consent.
C	Absolute language	Only that student's data must be excluded.
D	Correct	A minor's assent is needed in addition to parent permission.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	8	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Scholarly, peer-reviewed sources	Peer-reviewed journal articles have been checked by experts before publication and report methods, making them stronger evidence than blogs, magazines, or vendor pages.	Treating a popular website or trade magazine as scholarly because it cites research.	A choice among sources for a literature review.
2	Primary vs. secondary research sources	A primary source is the original study reporting its own data; a secondary source summarizes or discusses others' studies.	Citing a textbook or news summary as if it were the study itself.	Whether the authors collected the data they report.
3	Correlation is not causation	Two things moving together does not show one causes the other; a third factor or reverse direction may explain it.	Reading a strong correlation as proof of cause.	A relationship found in existing data followed by a causal claim.
4	Design strength for causal claims	Random assignment to treatment and control groups is the strongest basis for a causal claim; designs without a control group are weaker.	Believing a before-and-after gain in one group proves an intervention worked.	A claim that X caused Y and a description of the design.
5	Synthesis, not summary	A literature review is organized by themes and shows how studies agree, conflict, and leave gaps, rather than summarizing one article at a time.	Writing an annotated list of sources and calling it a review.	How a review is organized.
6	Purpose of action research	Action research is done by practitioners in their own setting to improve local practice through cycles of planning, acting, and reflecting; it does not aim to generalize.	Expecting classroom action research to prove results for all schools.	A teacher studying a change in her own classroom.
7	A researchable question	A good practitioner research question is specific, names what will be measured, and is within the researcher's control to study.	Asking a broad or unanswerable question such as how to fix all learning gaps.	A choice among draft research questions.
8	Consent and assent	Research with minors needs parent or guardian permission plus the child's own agreement (assent), participation must be voluntary, and data kept confidential.	Assuming a parent's permission alone is enough, or that routine class activities need no review when used as research data.	Collecting data from students for a study.

Course scope (public description): Educational Research focuses on practical problem solving. This course teaches candidates to use scholarly literature and current research to inform their own practice. It also empowers candidates to recognize opportunities for improvement and engage in action research to systematically implement and evaluate changes. This course prepares candidates to conduct research for the capstone. Data-Informed Practices is a prerequisite for this course.